

SDS8102: Advanced Mathematical Statistics

Lecture 13 & 14: Asymptotic properties of estimators

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Consistency and Asymptotic Normality of M- and Z-Estimators

- Consistency of M-estimators
- Consistency of Z-estimators
- Asymptotic normality and linearization
- Sandwich covariance
- Maximum likelihood estimation
- Correctly specified and misspecified models

From Empirical Processes to Estimation

Last week, we studied

$$\sup_{f \in \mathcal{F}} |P_n f - P f|.$$

The main message was:

control of complexity $\implies$ uniform convergence.

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$$\sup_{f \in \mathcal{F}} |P_n f - P f|.$$

The main message was:

control of complexity $\implies$ uniform convergence.

This week we ask:

How does uniform convergence help us understand estimators?

For M-estimators,

$$\hat{\theta}_n \in \arg \max_{\theta \in \Theta} P_n m_\theta.$$

Thus the function class is naturally

$$\mathcal{M} = \{m_\theta : \theta \in \Theta\}.$$

Two Fundamental Questions

Given an estimator $\hat{\theta}_n$ of θ_0 , we ask:

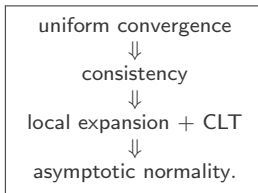
1. Is it consistent?

$$\hat{\theta}_n \xrightarrow{P} \theta_0.$$

2. What are its fluctuations around θ_0 ?

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} ?$$

The general strategy will be



M-Estimators

Let

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P.$$

For each $\theta \in \Theta$, let

$$m_\theta : \mathcal{X} \rightarrow \mathbb{R}.$$

Define

$$M_n(\theta) = P_n m_\theta = \frac{1}{n} \sum_{i=1}^n m_\theta(X_i),$$

and

$$M(\theta) = P m_\theta = \mathbb{E}[m_\theta(X)].$$

Definition:

An M-estimator satisfies

$$\hat{\theta}_n \in \arg \max_{\theta \in \Theta} M_n(\theta).$$

Examples of M-Estimators

Maximum likelihood:

$$m_{\theta}(x) = \log p_{\theta}(x).$$

Then

$$\hat{\theta}_{\text{MLE}} \in \arg \max_{\theta} \frac{1}{n} \sum_{i=1}^n \log p_{\theta}(X_i).$$

Least squares:

$$m_{\theta}(x, y) = -(y - x^{\top} \theta)^2.$$

Least absolute deviations:

$$m_{\theta}(x, y) = -|y - x^{\top} \theta|.$$

Remark:

Many apparently different estimators can therefore be studied using the same asymptotic theory.

The Population Objective

The empirical estimator maximizes

$$M_n(\theta).$$

Its population analogue is

$$M(\theta) = \mathbb{E}[m_\theta(X)].$$

Suppose

$$\theta_0 = \arg \max_{\theta \in \Theta} M(\theta).$$

If

$$M_n(\theta) \approx M(\theta)$$

uniformly over θ , we expect their maximizers to be close:

$M_n \approx M \implies \arg \max M_n \approx \arg \max M.$

Making this statement rigorous gives consistency of M-estimators.

Why Pointwise LLN Is Not Enough

For every fixed θ , the LLN gives

$$M_n(\theta) = P_n m_\theta \xrightarrow{P} P m_\theta = M(\theta).$$

But $\hat{\theta}_n$ is **not fixed**:

$$\hat{\theta}_n \in \arg \max_{\theta} M_n(\theta).$$

Pointwise convergence does not control

$$M_n(\hat{\theta}_n) - M(\hat{\theta}_n).$$

What we need is

$$\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \xrightarrow{P} 0.$$

This is exactly a uniform law of large numbers.

The Two Ingredients

Consistency requires two conceptually different ingredients.

1. Uniform convergence

$$\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \xrightarrow{P} 0.$$

2. Identification

For every $\varepsilon > 0$,

$$M(\theta_0) > \sup_{d(\theta, \theta_0) \geq \varepsilon} M(\theta).$$

Remark:

Uniform convergence controls the *randomness* of the criterion.
Identification controls the *geometry* of the population criterion.

Population Separation

For $\varepsilon > 0$, define

$$\Delta_\varepsilon = M(\theta_0) - \sup_{d(\theta, \theta_0) \geq \varepsilon} M(\theta).$$

Identification means

$$\Delta_\varepsilon > 0.$$

Suppose

$$\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| < \frac{\Delta_\varepsilon}{3}.$$

Then

$$M_n(\theta_0) > M(\theta_0) - \frac{\Delta_\varepsilon}{3},$$

whereas, for $d(\theta, \theta_0) \geq \varepsilon$,

$$M_n(\theta) < M(\theta_0) - \frac{2\Delta_\varepsilon}{3}.$$

Hence

$$M_n(\theta_0) > M_n(\theta)$$

for every θ outside the ε -neighborhood.

Consistency of M-Estimators

Theorem (Consistency of M-estimators)

Suppose

$$\hat{\theta}_n \in \arg \max_{\theta \in \Theta} M_n(\theta).$$

Assume:

1.

$$\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \xrightarrow{P} 0;$$

2. for every $\varepsilon > 0$,

$$\sup_{d(\theta, \theta_0) \geq \varepsilon} M(\theta) < M(\theta_0).$$

Then

$$\boxed{\hat{\theta}_n \xrightarrow{P} \theta_0.}$$

Proof

Let

$$D_n = \sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)|.$$

Suppose

$$d(\hat{\theta}_n, \theta_0) \geq \varepsilon.$$

Then

$$M(\hat{\theta}_n) \leq M(\theta_0) - \Delta_\varepsilon.$$

Since $\hat{\theta}_n$ maximizes M_n ,

$$M_n(\hat{\theta}_n) \geq M_n(\theta_0).$$

Therefore

$$M(\theta_0) - \Delta_\varepsilon + D_n \geq M_n(\hat{\theta}_n) \geq M_n(\theta_0) \geq M(\theta_0) - D_n.$$

Thus

$$\Delta_\varepsilon \leq 2D_n.$$

Hence

$$P\left(d(\hat{\theta}_n, \theta_0) \geq \varepsilon\right) \leq P\left(D_n \geq \frac{\Delta_\varepsilon}{2}\right) \rightarrow 0.$$

Where Last Week Enters

Since

$$M_n(\theta) - M(\theta) = (P_n - P)m_\theta,$$

we have

$$\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| = \sup_{m \in \mathcal{M}} |(P_n - P)m|,$$

where

$$\mathcal{M} = \{m_\theta : \theta \in \Theta\}.$$

Therefore,

$$\mathcal{M} \text{ is Glivenko-Cantelli} \implies \sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \rightarrow 0.$$

Consequently,

$$\text{Glivenko-Cantelli} + \text{Identification} \implies \text{Consistency}.$$

When Is the Criterion Class Glivenko–Cantelli?

Recall

$$\mathcal{M} = \{m_\theta : \theta \in \Theta\}.$$

Theorem

Suppose:

1. $\Theta \subset \mathbb{R}^d$ is compact;
2. for P -a.e. x ,

$$\theta \mapsto m_\theta(x)$$

is continuous;

3. there exists an envelope F such that

$$\sup_{\theta \in \Theta} |m_\theta(x)| \leq F(x), \quad PF < \infty.$$

Then

$$\sup_{\theta \in \Theta} |(P_n - P)m_\theta| \xrightarrow{P} 0.$$

Thus $\mathcal{M}$ is P -Glivenko–Cantelli.

Why? Local Oscillations Are Small

For $\theta \in \Theta$ and $\delta > 0$, define

$$G_{\theta,\delta}(x) = \sup_{\|\eta - \theta\| < \delta} |m_\eta(x) - m_\theta(x)|.$$

By continuity,

$$G_{\theta,\delta}(x) \downarrow 0 \quad \text{as } \delta \downarrow 0$$

for P -a.e. x .

Moreover,

$$G_{\theta,\delta}(x) \leq 2F(x), \quad PF < \infty.$$

Hence, by dominated convergence,

$$PG_{\theta,\delta} \longrightarrow 0 \quad \text{as } \delta \downarrow 0.$$

Thus, for every $\varepsilon > 0$ and every θ , we can find a neighborhood U_θ such that

$$P \left[\sup_{\eta \in U_\theta} |m_\eta - m_\theta| \right] < \varepsilon.$$

Proof of the Glivenko–Cantelli Property

By compactness, finitely many such neighborhoods cover Θ :

$$\Theta \subset U_{\theta_1} \cup \dots \cup U_{\theta_K}.$$

For $\eta \in U_{\theta_j}$,

$$\begin{aligned} |(P_n - P)m_\eta| &\leq |(P_n - P)m_{\theta_j}| \\ &\quad + P_n|m_\eta - m_{\theta_j}| + P|m_\eta - m_{\theta_j}|. \end{aligned}$$

Therefore

$$\begin{aligned} \sup_{\eta \in \Theta} |(P_n - P)m_\eta| &\leq \max_{j \leq K} |(P_n - P)m_{\theta_j}| \\ &\quad + \max_{j \leq K} P_n G_j + \max_{j \leq K} P G_j, \end{aligned}$$

where

$$G_j = \sup_{\eta \in U_{\theta_j}} |m_\eta - m_{\theta_j}|.$$

By the LLN,

$$\max_{j \leq K} |(P_n - P)m_{\theta_j}| \rightarrow 0, \quad P_n G_j \rightarrow P G_j.$$

Since $P G_j < \varepsilon$,

$$\limsup_n \sup_{\theta \in \Theta} |(P_n - P)m_\theta| \leq 2\varepsilon$$

A Convenient Lipschitz Condition

A simpler condition is often easier to verify.

Suppose $\Theta \subset \mathbb{R}^d$ is compact and

$$|m_\theta(x) - m_\eta(x)| \leq L(x)\|\theta - \eta\|, \quad PL < \infty.$$

Also suppose

$$P|m_{\theta_*}| < \infty$$

for some $\theta_* \in \Theta$.

Then

$$\mathcal{M} = \{m_\theta : \theta \in \Theta\}$$

is P -Glivenko–Cantelli.

Indeed, take a finite δ -net

$$\{\theta_1, \dots, \theta_N\}$$

of Θ . For every θ , choose θ_j such that

$$\|\theta - \theta_j\| \leq \delta.$$

Then

$$\sup_{\theta \in \Theta} |(P_n - P)m_\theta| \leq \max_{j \leq N} |(P_n - P)m_{\theta_j}| + \delta(P_n L + PL).$$

Proof Under the Lipschitz Condition

For fixed δ , the net is finite. Hence the LLN gives

$$\max_{j \leq N} |(P_n - P)m_{\theta_j}| \xrightarrow{P} 0.$$

Also,

$$P_n L \xrightarrow{P} PL.$$

Therefore

$$\limsup_{n \rightarrow \infty} \sup_{\theta \in \Theta} |(P_n - P)m_{\theta}| \leq 2\delta PL$$

in probability.

Since $\delta > 0$ is arbitrary,

$$\sup_{\theta \in \Theta} |(P_n - P)m_{\theta}| \xrightarrow{P} 0.$$

Remark:

Compactness gives a *finite net*; Lipschitz continuity controls what happens between the net points.

Z-Estimators

Instead of optimizing an objective function, many estimators are defined by solving an *estimating equation*.

Let

$$\psi_{\theta} : \mathcal{X} \rightarrow \mathbb{R}^k.$$

Define

$$\Psi_n(\theta) = P_n \psi_{\theta} = \frac{1}{n} \sum_{i=1}^n \psi_{\theta}(X_i),$$

and

$$\Psi(\theta) = P \psi_{\theta} = \mathbb{E}[\psi_{\theta}(X)].$$

Definition:

A Z-estimator $\hat{\theta}_n$ satisfies

$$\Psi_n(\hat{\theta}_n) = 0.$$

The population parameter satisfies

$$\Psi(\theta_0) = 0.$$

Examples of Z-Estimators

Method of moments:

$$\psi_{\theta}(x) = g(x) - \mu(\theta).$$

Then

$$P_n g = \mu(\hat{\theta}_n).$$

Maximum likelihood:

If

$$\ell_{\theta}(x) = \log p_{\theta}(x) \Rightarrow \psi_{\theta}(x) = \nabla_{\theta} \ell_{\theta}(x).$$

An interior MLE satisfies

$$P_n \psi_{\hat{\theta}_n} = 0.$$

Regression:

$$\psi_{\theta}(x, y) = x(y - x^{\top} \theta).$$

Then

$$P_n \psi_{\hat{\theta}_n} = 0$$

gives the normal equations for least squares.

M-Estimators versus Z-Estimators

The two frameworks look slightly different.

M-estimation	Z-estimation
$\hat{\theta}_n \in \arg \max_{\theta} M_n(\theta)$	$\Psi_n(\hat{\theta}_n) = 0$
$\theta_0 \in \arg \max_{\theta} M(\theta)$	$\Psi(\theta_0) = 0$

When M_n is differentiable,

$$\Psi_n(\theta) = \nabla_{\theta} M_n(\theta),$$

so an interior M-estimator is also a Z-estimator.

optimization $\longleftrightarrow$ estimating equations.
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What Should Give Consistency?

Suppose

$$\Psi_n(\hat{\theta}_n) = 0.$$

If

$$\sup_{\theta \in \Theta} \|\Psi_n(\theta) - \Psi(\theta)\| \xrightarrow{P} 0,$$

then

$$\Psi(\hat{\theta}_n)$$

should be close to zero.

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$$\Psi(\hat{\theta}_n)$$

should be close to zero.

But this alone is not enough.

We also need the population equation

$$\Psi(\theta) = 0$$

to identify θ_0 .

For every $\varepsilon > 0$, require

$$\inf_{d(\theta, \theta_0) \geq \varepsilon} \|\Psi(\theta)\| > 0.$$

Thus $\Psi(\theta)$ is bounded away from zero away from θ_0 .

Consistency of Z-Estimators

Theorem (Consistency of Z-estimators)

Suppose

$$\|\Psi_n(\hat{\theta}_n)\| \xrightarrow{P} 0.$$

Assume:

1.

$$\sup_{\theta \in \Theta} \|\Psi_n(\theta) - \Psi(\theta)\| \xrightarrow{P} 0;$$

2. for every $\varepsilon > 0$,

$$\inf_{d(\theta, \theta_0) \geq \varepsilon} \|\Psi(\theta)\| > 0.$$

Then

$$\boxed{\hat{\theta}_n \xrightarrow{P} \theta_0.}$$

Proof

Fix $\varepsilon > 0$ and define

$$c_\varepsilon = \inf_{d(\theta, \theta_0) \geq \varepsilon} \|\Psi(\theta)\| > 0.$$

If

$$d(\hat{\theta}_n, \theta_0) \geq \varepsilon,$$

then

$$c_\varepsilon \leq \|\Psi(\hat{\theta}_n)\|.$$

By the triangle inequality,

$$\begin{aligned} \|\Psi(\hat{\theta}_n)\| &\leq \|\Psi(\hat{\theta}_n) - \Psi_n(\hat{\theta}_n)\| + \|\Psi_n(\hat{\theta}_n)\| \\ &\leq \sup_{\theta \in \Theta} \|\Psi_n(\theta) - \Psi(\theta)\| + \|\Psi_n(\hat{\theta}_n)\|. \end{aligned}$$

The right-hand side is $o_P(1)$.

Therefore

$$P\left(d(\hat{\theta}_n, \theta_0) \geq \varepsilon\right) \rightarrow 0.$$

M- and Z-Consistency: The Same Principle

The two consistency results have the same structure.

empirical object uniformly approximates population object

+

population problem uniquely identifies θ_0

$\Downarrow$

$$\hat{\theta}_n \xrightarrow{P} \theta_0.$$

For M-estimators:

$$M_n(\theta) \approx M(\theta).$$

For Z-estimators:

$$\Psi_n(\theta) \approx \Psi(\theta).$$

Remark:

Consistency is a **global** property: we must rule out parameters that are far away from θ_0 .

Consistency Is Only the First Step

Consistency tells us

$$\hat{\theta}_n - \theta_0 = o_P(1).$$

But for statistical inference, we need to know:

How fast does $\hat{\theta}_n$ converge?

and

What is the distribution of its estimation error?

Typically,

$$\hat{\theta}_n - \theta_0 = O_P(n^{-1/2}),$$

and we seek a limit for

$$\sqrt{n}(\hat{\theta}_n - \theta_0).$$

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The key idea is to **linearize the estimating equation** around θ_0 .

The Basic Idea

Suppose, for simplicity, $\theta \in \mathbb{R}$ and

$$\Psi_n(\widehat{\theta}_n) = 0.$$

Taylor expansion around θ_0 gives

$$0 = \Psi_n(\theta_0) + \dot{\Psi}_n(\widetilde{\theta}_n)(\widehat{\theta}_n - \theta_0),$$

where $\widetilde{\theta}_n$ lies between $\widehat{\theta}_n$ and θ_0 .

Therefore

$$\widehat{\theta}_n - \theta_0 = -\frac{\Psi_n(\theta_0)}{\dot{\Psi}_n(\widetilde{\theta}_n)}.$$

Multiplying by $\sqrt{n}$,

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = -\frac{\sqrt{n}\Psi_n(\theta_0)}{\dot{\Psi}_n(\widetilde{\theta}_n)}.$$

Now numerator and denominator can be studied separately.

The Numerator: A Central Limit Theorem

Since

$$\Psi(\theta_0) = \mathbb{E}[\psi_{\theta_0}(X)] = 0,$$

we have

$$\sqrt{n} \Psi_n(\theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_{\theta_0}(X_i).$$

Therefore, under finite second moments,

$$\sqrt{n} \Psi_n(\theta_0) \xrightarrow{d} N(0, B),$$

where

$$B = \text{Var}(\psi_{\theta_0}(X)).$$

This is just an ordinary CLT.

The Denominator: Local Stability

Suppose

$$\dot{\Psi}_n(\tilde{\theta}_n) \xrightarrow{P} A,$$

where

$$A = \dot{\Psi}(\theta_0) \neq 0.$$

Then

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -A^{-1}\sqrt{n}\Psi_n(\theta_0) + o_P(1).$$

By the CLT and Slutsky's theorem,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N\left(0, \frac{B}{A^2}\right).$$

Remark:

Consistency localizes $\hat{\theta}_n$ near θ_0 .

Differentiability then turns the nonlinear estimating equation into a linear one.

The Multivariate Case

Now let

$$\theta \in \mathbb{R}^d, \quad \psi_\theta(X) \in \mathbb{R}^d.$$

Recall

$$\Psi_n(\theta) = P_n \psi_\theta, \quad \Psi(\theta) = P \psi_\theta.$$

Suppose

$$\Psi(\theta_0) = 0, \quad \Psi_n(\hat{\theta}_n) = 0.$$

The derivative is now the $d \times d$ Jacobian

$$A = \dot{\Psi}(\theta_0) = \nabla_\theta \Psi(\theta)|_{\theta=\theta_0}.$$

We assume that A is nonsingular.

Linearizing the Estimating Equation

Since

$$\Psi_n(\widehat{\theta}_n) = 0,$$

expand around θ_0 :

$$0 = \Psi_n(\theta_0) + \dot{\Psi}_n(\widetilde{\theta}_n)(\widehat{\theta}_n - \theta_0),$$

where $\widetilde{\theta}_n$ lies between $\widehat{\theta}_n$ and θ_0 .

Hence

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = -\dot{\Psi}_n(\widetilde{\theta}_n)^{-1} \sqrt{n} \Psi_n(\theta_0).$$

If

$$\dot{\Psi}_n(\widetilde{\theta}_n) \xrightarrow{P} A,$$

then

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = -A^{-1} \sqrt{n} \Psi_n(\theta_0) + o_P(1).$$

Asymptotic Linear Representation

Since

$$\Psi(\theta_0) = \mathbb{E}[\psi_{\theta_0}(X)] = 0,$$

we have

$$\sqrt{n} \Psi_n(\theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_{\theta_0}(X_i).$$

Therefore

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \phi(X_i) + o_P(1),$$

where

$$\phi(x) = -A^{-1} \psi_{\theta_0}(x).$$

Definition:

A representation of this form is called an *asymptotic linear representation*. The function ϕ is called the *influence function*.

Why Asymptotic Linearity Is Powerful

The estimator itself may be a complicated nonlinear function of the entire sample.

But asymptotically,

$$\hat{\theta}_n - \theta_0 = \frac{1}{n} \sum_{i=1}^n \phi(X_i) + o_P(n^{-1/2}).$$

Thus, to first order,

complicated estimator $\approx$ average of i.i.d. random variables.

Once we have this representation,

- the CLT gives asymptotic normality;
- the variance is determined by ϕ ;
- standard errors can be estimated;
- confidence intervals can be constructed.

Asymptotic Normality of Z-Estimators

Define

$$B = \text{Var}(\psi_{\theta_0}(X)).$$

By the multivariate CLT,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_{\theta_0}(X_i) \xrightarrow{d} N(0, B).$$

Therefore

Theorem (Asymptotic normality)

Under suitable regularity conditions,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, A^{-1}BA^{-T}).$$

The matrix

$$A^{-1}BA^{-T}$$

is called the *sandwich covariance matrix*.

The Sandwich Covariance

The asymptotic covariance is

$$V = A^{-1}BA^{-T}.$$

It contains two different quantities.

Variability of the estimating equation:

$$B = \text{Var}(\psi_{\theta_0}(X)).$$

Sensitivity of the population equation:

$$A = \left. \frac{\partial}{\partial \theta^T} \mathbb{E}[\psi_{\theta}(X)] \right|_{\theta=\theta_0}.$$

Remark:

Large B means noisy estimating equations.

A nearly singular A means that the population equation changes very slowly with θ , making θ_0 difficult to estimate.

A General Z-Estimator Theorem

Theorem

Suppose:

1.

$$\hat{\theta}_n \xrightarrow{P} \theta_0;$$

2.

$$\Psi_n(\hat{\theta}_n) = 0;$$

3. Ψ is differentiable at θ_0 with nonsingular

$$A = \dot{\Psi}(\theta_0);$$

4. locally around θ_0 ,

$$\dot{\Psi}_n(\theta) \xrightarrow{P} \dot{\Psi}(\theta)$$

sufficiently uniformly;

5.

$$\sqrt{n} \Psi_n(\theta_0) \xrightarrow{d} N(0, B).$$

Then

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, A^{-1}BA^{-T}).$$

The Logic of the Proof

The argument has four steps.

$$\begin{array}{c} \hat{\theta}_n \rightarrow \theta_0 \\ \Downarrow \\ \text{Taylor expansion around } \theta_0 \\ \Downarrow \\ \sqrt{n}(\hat{\theta}_n - \theta_0) = -A^{-1}\sqrt{n}\Psi_n(\theta_0) + o_P(1) \\ \Downarrow \\ \text{CLT} + \text{Slutsky} \\ \Downarrow \\ N(0, A^{-1}BA^{-T}). \end{array}$$

Remark:

Consistency is what allows us to replace a *global* estimation problem by a *local* linear approximation around θ_0 .

Example: Estimating the Mean

Let

$$\theta_0 = \mathbb{E}[X].$$

Take

$$\psi_\theta(x) = x - \theta.$$

The estimating equation is

$$0 = P_n(X - \hat{\theta}_n),$$

so

$$\hat{\theta}_n = \bar{X}_n.$$

The population equation is

$$\Psi(\theta) = \mathbb{E}[X] - \theta = \theta_0 - \theta.$$

Therefore

$$A = \dot{\Psi}(\theta_0) = -1,$$

and

$$B = \text{Var}(X) = \sigma^2.$$

Hence

$$\sqrt{n}(\bar{X}_n - \theta_0) \Rightarrow N(0, \sigma^2).$$

So the ordinary CLT is itself a special case of Z-estimation theory.

Example: Method of Moments

Suppose

$$\mathbb{E}[g(X)] = \mu(\theta_0),$$

where

$$\mu : \mathbb{R}^d \rightarrow \mathbb{R}^d.$$

Take

$$\psi_\theta(x) = g(x) - \mu(\theta).$$

Then

$$\Psi_n(\hat{\theta}_n) = 0$$

means

$$P_n g = \mu(\hat{\theta}_n).$$

Moreover,

$$A = -\dot{\mu}(\theta_0),$$

and

$$B = \text{Var}(g(X)).$$

Therefore

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N\left(0, \dot{\mu}(\theta_0)^{-1} B \dot{\mu}(\theta_0)^{-T}\right).$$

Now Return to Maximum Likelihood

For a parametric model

$$\{p_\theta : \theta \in \Theta\},$$

define the log-likelihood contribution

$$\ell_\theta(x) = \log p_\theta(x).$$

The MLE is

$$\hat{\theta}_n \in \arg \max_{\theta \in \Theta} P_n \ell_\theta.$$

If $\hat{\theta}_n$ is an interior maximizer, then

$$P_n \dot{\ell}_{\hat{\theta}_n} = 0.$$

Thus the MLE is also a Z-estimator with

$$\psi_\theta(x) = \dot{\ell}_\theta(x) = \nabla_\theta \log p_\theta(x).$$

This function is the *score*.

The Score Has Mean Zero

Let

$$s_{\theta}(x) = \nabla_{\theta} \log p_{\theta}(x).$$

Under regularity conditions,

$$\begin{aligned}\mathbb{E}_{\theta}[s_{\theta}(X)] &= \int \nabla_{\theta} \log p_{\theta}(x) p_{\theta}(x) dx \\ &= \int \nabla_{\theta} p_{\theta}(x) dx \\ &= \nabla_{\theta} \int p_{\theta}(x) dx \\ &= 0.\end{aligned}$$

Therefore

$$\boxed{\mathbb{E}_{\theta_0}[s_{\theta_0}(X)] = 0.}$$

So the population score equation is correctly centered at the true parameter.

Fisher Information

The covariance of the score is

$$I(\theta) = \mathbb{E}_{\theta} \left[s_{\theta}(X) s_{\theta}(X)^{\top} \right].$$

Since

$$\mathbb{E}_{\theta}[s_{\theta}(X)] = 0,$$

this is also

$$I(\theta) = \text{Var}_{\theta} (s_{\theta}(X)).$$

Thus, in the general Z-estimator notation,

$$B = I(\theta_0).$$

What about the other matrix,

$$A = \mathbb{E}_{\theta_0} [\nabla_{\theta} s_{\theta_0}(X)]?$$

The Information Identity

Recall the score

$$s_{\theta}(x) = \nabla_{\theta} \log p_{\theta}(x).$$

We know

$$\mathbb{E}_{\theta}[s_{\theta}(X)] = 0.$$

Differentiate this identity with respect to θ :

$$0 = \nabla_{\theta} \int s_{\theta}(x) p_{\theta}(x) dx.$$

Using

$$\nabla_{\theta} p_{\theta}(x) = p_{\theta}(x) s_{\theta}(x),$$

we obtain

$$0 = \mathbb{E}_{\theta} \left[\nabla_{\theta} s_{\theta}(X) + s_{\theta}(X) s_{\theta}(X)^{\top} \right].$$

Therefore

$$\boxed{-\mathbb{E}_{\theta}[\nabla_{\theta}^2 \ell_{\theta}(X)] = \mathbb{E}_{\theta}[s_{\theta}(X) s_{\theta}(X)^{\top}] = I(\theta).}$$

MLE as a Z-Estimator

For the MLE,

$$\psi_{\theta}(x) = s_{\theta}(x).$$

Recall the general Z-estimator covariance:

$$A^{-1}BA^{-T},$$

where

$$A = \mathbb{E}_{\theta_0}[\nabla_{\theta} s_{\theta_0}(X)],$$

and

$$B = \text{Var}_{\theta_0}[s_{\theta_0}(X)].$$

By the information identity,

$$A = -I(\theta_0), \quad B = I(\theta_0).$$

Therefore

$$A^{-1}BA^{-T} = I(\theta_0)^{-1}.$$

Asymptotic Normality of the MLE

Under suitable regularity conditions,

$$\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta_0) \xrightarrow{d} N(0, I(\theta_0)^{-1}).$$

Equivalently, for large n ,

$$\hat{\theta}_{\text{MLE}} \approx N\left(\theta_0, \frac{1}{n} I(\theta_0)^{-1}\right).$$

Remark:

The Fisher information determines the local precision with which θ_0 can be estimated. Large information means small asymptotic variance.

The MLE Expansion

The score equation is

$$P_n s_{\hat{\theta}_n} = 0.$$

Linearizing around θ_0 ,

$$0 = P_n s_{\theta_0} + P_n \nabla_{\theta} s_{\theta_0} (\hat{\theta}_n - \theta_0) + o_P(n^{-1/2}).$$

Since

$$P_n \nabla_{\theta} s_{\theta_0} \xrightarrow{P} -I(\theta_0),$$

we obtain

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = I(\theta_0)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n s_{\theta_0}(X_i) + o_P(1).$$

Thus the influence function of the MLE is

$$\phi(x) = I(\theta_0)^{-1} s_{\theta_0}(x).$$

Example: Gaussian Mean

Suppose

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2),$$

where σ^2 is known.

For one observation,

$$\ell_\mu(x) = -\frac{(x - \mu)^2}{2\sigma^2} + \text{constant}.$$

The score is

$$s_\mu(x) = \frac{x - \mu}{\sigma^2}.$$

Hence

$$I(\mu) = \mathbb{E}_\mu \left[\frac{(X - \mu)^2}{\sigma^4} \right] = \frac{1}{\sigma^2}.$$

Therefore

$$\sqrt{n}(\hat{\mu} - \mu) \Rightarrow N(0, \sigma^2).$$

Of course,

$$\hat{\mu} = \bar{X}.$$

Example: Bernoulli MLE

Suppose

$$X_i \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p).$$

For one observation,

$$\ell_p(x) = x \log p + (1 - x) \log(1 - p).$$

The score is

$$s_p(x) = \frac{x}{p} - \frac{1-x}{1-p} = \frac{x-p}{p(1-p)}.$$

Thus

$$I(p) = \mathbb{E}_p[s_p(X)^2] = \frac{1}{p(1-p)}.$$

Hence

$$\boxed{\sqrt{n}(\hat{p} - p) \Rightarrow N(0, p(1-p))}.$$

Since $\hat{p} = \overline{X}$, this agrees with the ordinary CLT.

Estimating the Asymptotic Variance

The asymptotic result is

$$\hat{\theta}_n \approx N\left(\theta_0, \frac{1}{n} I(\theta_0)^{-1}\right).$$

But θ_0 is unknown.

A natural plug-in estimator is

$$I(\hat{\theta}_n).$$

Thus

$$\widehat{\text{Var}}(\hat{\theta}_n) = \frac{1}{n} I(\hat{\theta}_n)^{-1}.$$

Alternatively, use the observed information:

$$\hat{l}_n = -\frac{1}{n} \sum_{i=1}^n \nabla_{\hat{\theta}_n}^2 \ell_{\hat{\theta}_n}(X_i).$$

Then

$$\widehat{\text{Var}}(\hat{\theta}_n) = \frac{1}{n} \hat{l}_n^{-1}.$$

Wald Confidence Intervals

For a scalar parameter,

$$\frac{\sqrt{n}(\hat{\theta}_n - \theta_0)}{\sqrt{I(\theta_0)^{-1}}} \Rightarrow N(0, 1).$$

Replacing θ_0 by $\hat{\theta}_n$ gives the approximate $(1 - \alpha)$ confidence interval

$$\hat{\theta}_n \pm z_{1-\alpha/2} \sqrt{\frac{I(\hat{\theta}_n)^{-1}}{n}}.$$

More generally, for a smooth function

$$g : \mathbb{R}^d \rightarrow \mathbb{R},$$

the delta method gives

$$\sqrt{n}\{g(\hat{\theta}_n) - g(\theta_0)\} \Rightarrow N\left(0, \nabla g(\theta_0)^\top I(\theta_0)^{-1} \nabla g(\theta_0)\right).$$

What If the Model Is Wrong?

So far we assumed

$$P_0 = P_{\theta_0}$$

for some $\theta_0 \in \Theta$.

But in practice, the true distribution P_0 may not belong to the parametric model

$$\{P_\theta : \theta \in \Theta\}.$$

The MLE still solves

$$\hat{\theta}_n \in \arg \max_{\theta \in \Theta} P_n \log p_\theta.$$

What does it converge to?

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$$\hat{\theta}_n \in \arg \max_{\theta \in \Theta} P_n \log p_\theta.$$

What does it converge to?

The population objective is now

$$M(\theta) = P_0 \log p_\theta.$$

Thus the target becomes

$$\theta^* = \arg \max_{\theta \in \Theta} P_0 \log p_\theta.$$

The KL Projection

Observe that

$$\begin{aligned} D_{\text{KL}}(P_0 \| P_\theta) &= P_0 \log \frac{p_0}{p_\theta} \\ &= P_0 \log p_0 - P_0 \log p_\theta. \end{aligned}$$

The first term does not depend on θ .

Therefore

$$\theta^* = \arg \min_{\theta \in \Theta} D_{\text{KL}}(P_0 \| P_\theta).$$

So even under misspecification, the MLE has a meaningful population target:

the model distribution closest to the truth in KL divergence.

This parameter is sometimes called the *pseudo-true parameter*.

Consistency Under Misspecification

The same M-estimation argument applies.

If

$$\sup_{\theta \in \Theta} |P_n \log p_\theta - P_0 \log p_\theta| \xrightarrow{P} 0,$$

and θ^* is suitably identified, then

$$\hat{\theta}_{\text{MLE}} \xrightarrow{P} \theta^*.$$

Remark:

Consistency does not necessarily mean convergence to a “true” parameter.
It means convergence to the population target defined by the statistical procedure.

What Happens to the Information Identity?

At θ^* , define

$$s_{\theta^*}(x) = \nabla_{\theta} \log p_{\theta^*}(x).$$

The first-order condition gives

$$P_0 s_{\theta^*} = 0.$$

Define

$$A = P_0 [\nabla_{\theta} s_{\theta^*}(X)],$$

and

$$B = P_0 [s_{\theta^*}(X) s_{\theta^*}(X)^{\top}].$$

Under correct specification,

$$A = -B.$$

Under misspecification, however,

$-A \neq B$ in general.

The information identity no longer holds.

The Sandwich Returns

The general Z-estimator theorem gives

$$\sqrt{n}(\hat{\theta}_n - \theta^*) \Rightarrow N(0, A^{-1}BA^{-T}).$$

Thus the asymptotic covariance is again the full sandwich.

$$\underbrace{A^{-1}}_{\text{bread}} \underbrace{B}_{\text{meat}} \underbrace{A^{-T}}_{\text{bread}}$$

Under correct specification,

$$A = -I(\theta_0), \quad B = I(\theta_0),$$

and the sandwich simplifies to

$$I(\theta_0)^{-1}.$$

Under misspecification, we must keep the full sandwich covariance.

Estimating the Sandwich Covariance

Define

$$\hat{A} = \frac{1}{n} \sum_{i=1}^n \nabla_{\theta} s_{\hat{\theta}_n}(X_i),$$

and

$$\hat{B} = \frac{1}{n} \sum_{i=1}^n s_{\hat{\theta}_n}(X_i) s_{\hat{\theta}_n}(X_i)^{\top}.$$

Then

$$\hat{V} = \hat{A}^{-1} \hat{B} \hat{A}^{-\top}.$$

The estimated covariance of $\hat{\theta}_n$ is

$$\widehat{\text{Var}}(\hat{\theta}_n) = \frac{1}{n} \hat{V}.$$

Remark:

This is the basic idea behind *sandwich* or *robust* standard errors.

Correct Specification versus Misspecification

	Correct model	Misspecified model
Target	θ_0	θ^*
Population meaning	$P_0 = P_{\theta_0}$	$\min_{\theta} D_{\text{KL}}(P_0 \ P_{\theta})$
A	$-I(\theta_0)$	$P_0 \nabla s_{\theta^*}$
B	$I(\theta_0)$	$P_0 s_{\theta^*} s_{\theta^*}^{\top}$
Asymptotic variance	$I(\theta_0)^{-1}$	$A^{-1} B A^{-\top}$

Fisher information is a special simplification of the sandwich.

Back to General M-Estimators

Recall

$$\hat{\theta}_n \in \arg \max_{\theta \in \Theta} M_n(\theta), \quad M_n(\theta) = P_n m_\theta.$$

We already showed that, under uniform convergence and identification,

$$\hat{\theta}_n \xrightarrow{P} \theta_0.$$

Now suppose m_θ is differentiable in θ .

For an interior maximizer,

$$\nabla_\theta M_n(\hat{\theta}_n) = 0.$$

Since

$$\nabla_\theta M_n(\theta) = P_n \nabla_\theta m_\theta,$$

a smooth M-estimator is a Z-estimator with

$$\psi_\theta(x) = \nabla_\theta m_\theta(x).$$

Thus the Z-estimator theory immediately gives asymptotic normality.

Asymptotic Normality of Smooth M-Estimators

Let

$$\dot{m}_{\theta}(x) = \nabla_{\theta} m_{\theta}(x), \quad \ddot{m}_{\theta}(x) = \nabla_{\theta}^2 m_{\theta}(x).$$

Define

$$A = P \ddot{m}_{\theta_0},$$

and

$$B = \text{Var}(\dot{m}_{\theta_0}(X)).$$

Under suitable regularity conditions,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N(0, A^{-1}BA^{-T}).$$

The corresponding asymptotic linear representation is

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -A^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \dot{m}_{\theta_0}(X_i) + o_P(1).$$

An Alternative View: Quadratic Approximation

There is another way to understand the same result.

Expand the empirical objective around θ_0 :

$$\begin{aligned} M_n(\theta) &\approx M_n(\theta_0) + (\theta - \theta_0)^\top \nabla M_n(\theta_0) \\ &\quad + \frac{1}{2}(\theta - \theta_0)^\top \nabla^2 M_n(\theta_0)(\theta - \theta_0). \end{aligned}$$

The first term does not affect the maximizer.

The second term is random:

$$\nabla M_n(\theta_0) = P_n \dot{m}_{\theta_0} = O_P(n^{-1/2}).$$

The quadratic term determines the local curvature.

Maximizing the Quadratic Approximation

The first-order condition for the quadratic approximation is

$$0 = \nabla M_n(\theta_0) + \nabla^2 M_n(\theta_0)(\hat{\theta}_n - \theta_0).$$

Hence

$$\hat{\theta}_n - \theta_0 \approx -\{\nabla^2 M_n(\theta_0)\}^{-1} \nabla M_n(\theta_0).$$

Since

$$\nabla^2 M_n(\theta_0) \xrightarrow{P} A,$$

we obtain

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \approx -A^{-1} \sqrt{n} P_n \dot{m}_{\theta_0}.$$

So asymptotic normality comes from

$$\text{random first-order fluctuation} \quad + \quad \text{deterministic local curvature.}$$

Global versus Local Arguments

It is useful to separate two parts of asymptotic theory.

Consistency	Asymptotic normality
global	local
uniform convergence	Taylor expansion
identification	local curvature
$\hat{\theta}_n \rightarrow \theta_0$	$\sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N(0, V)$

Remark:

Consistency gets us into a small neighborhood of θ_0 .
Once we are there, local smoothness determines the first-order distribution.

Why Does the $\sqrt{n}$ Rate Appear?

Consider again

$$0 \approx \Psi_n(\theta_0) + A(\hat{\theta}_n - \theta_0).$$

Therefore

$$\hat{\theta}_n - \theta_0 \approx -A^{-1}\Psi_n(\theta_0).$$

But

$$\Psi_n(\theta_0) = \frac{1}{n} \sum_{i=1}^n \psi_{\theta_0}(X_i),$$

where

$$\mathbb{E}[\psi_{\theta_0}(X)] = 0.$$

By the CLT,

$$\Psi_n(\theta_0) = O_P(n^{-1/2}).$$

Hence

$$\boxed{\hat{\theta}_n - \theta_0 = O_P(n^{-1/2}).}$$

The root- n rate is inherited directly from averaging i.i.d. noise.

When Can the Root- n Theory Fail?

The previous argument relies on local regularity.

The usual root- n normal approximation may fail if:

- the parameter is on the boundary;
- the population objective is not sufficiently smooth;
- the local curvature is singular;
- the parameter is not locally identified;
- the criterion has nonstandard behavior near θ_0 .

For example, if

$$A = \dot{\Psi}(\theta_0)$$

is singular, we cannot write

$$\hat{\theta}_n - \theta_0 \approx -A^{-1}\Psi_n(\theta_0).$$

Remark:

Consistency alone does not imply either a root- n rate or asymptotic normality.

From Asymptotic Normality to Inference

Suppose

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N(0, V).$$

If

$$\hat{V} \xrightarrow{P} V,$$

then

$$\hat{V}^{-1/2} \sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N(0, I_d).$$

For the j th coordinate,

$$\frac{\hat{\theta}_{n,j} - \theta_{0,j}}{\sqrt{\hat{V}_{jj}/n}} \Rightarrow N(0, 1).$$

Hence an approximate confidence interval is

$$\hat{\theta}_{n,j} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{V}_{jj}}{n}}.$$

Estimating the Sandwich Variance

For a general Z-estimator, define

$$\hat{A} = \frac{1}{n} \sum_{i=1}^n \nabla_{\theta} \psi_{\hat{\theta}_n}(X_i),$$

and

$$\hat{B} = \frac{1}{n} \sum_{i=1}^n \psi_{\hat{\theta}_n}(X_i) \psi_{\hat{\theta}_n}(X_i)^{\top}.$$

Then

$$\hat{A} \xrightarrow{P} A, \quad \hat{B} \xrightarrow{P} B.$$

Thus

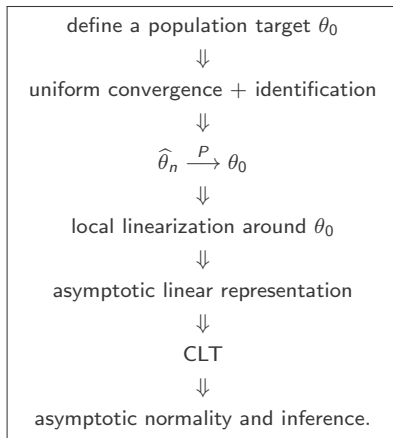
$$\boxed{\hat{V} = \hat{A}^{-1} \hat{B} \hat{A}^{-\top}.$$

The estimated covariance of $\hat{\theta}_n$ is

$$\frac{\hat{V}}{n}.$$

The General M/Z-Estimation Pipeline

Many estimation problems can be understood through the same sequence:



How the Pieces Fit Together

	M-estimator	Z-estimator
Empirical problem	$\max_{\theta} M_n(\theta)$	$\Psi_n(\theta) = 0$
Population problem	$\max_{\theta} M(\theta)$	$\Psi(\theta) = 0$
Consistency	$M_n \approx M$	$\Psi_n \approx \Psi$
Local object	∇M_n	Ψ_n
Derivative	$P\ddot{m}_{\theta_0}$	$\dot{\Psi}(\theta_0)$
Limit variance	$A^{-1}BA^{-T}$	$A^{-1}BA^{-T}$

For smooth problems, the two frameworks are essentially the same.

Maximum Likelihood: The Complete Picture

For maximum likelihood,

$$m_{\theta}(x) = \log p_{\theta}(x).$$

Population criterion:

$$P_{\theta_0} \log p_{\theta}.$$

Identification:

$$M(\theta_0) - M(\theta) = D_{\text{KL}}(P_{\theta_0} \| P_{\theta}).$$

Estimating equation:

$$P_n s_{\hat{\theta}_n} = 0.$$

Local curvature:

$$-P_{\theta_0} \nabla^2 \ell_{\theta_0} = I(\theta_0).$$

Asymptotic distribution:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N(0, I(\theta_0)^{-1}).$$

Connection to Last Week

Last week:

complexity $\implies$ uniform LLN / CLT.

This week:

uniform LLN $\implies$ consistency,

and

local empirical fluctuations + linearization $\implies$ asymptotic normality.

Together,

empirical process theory $\implies$ large-sample theory of estimators.

Takeaways

- M-estimators optimize an empirical criterion.
- Z-estimators solve empirical estimating equations.
- Uniform convergence plus identification gives consistency.
- Consistency allows us to localize around the population parameter.
- Linearization gives an asymptotic linear representation.
- The general asymptotic covariance is

$$A^{-1}BA^{-T}.$$

- For a correctly specified likelihood, the information identity simplifies this to

$$I(\theta_0)^{-1}.$$

- Under misspecification, the full sandwich covariance remains.